

Identifying the closest reference sequence in long-read fungal community profiling

Rationale, experimental validation and limitations of the TaxQuality development phase
(LOREON pipeline)

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1 Introduction: the question addressed by this development phase

In reference-based profiling of microbial communities, every sequencing read is compared against an archive of reference sequences and assigned to the one it most closely resembles. The outcome of this comparison — conventionally termed the *primary* assignment — constitutes the elementary datum upon which all downstream estimates rest: relative abundances, species richness and diversity indices.

The present phase of work originates from a simple observation and from the question that follows from it.

The observation. The primary assignment is not a measurement but a **choice**. When a read resembles several reference sequences almost equally — the ordinary situation for the ITS marker within archives such as UNITE, which contain thousands of closely related species — the alignment algorithm designates one of them, and the remainder disappear from the result. That choice is governed by score differences that are frequently negligible and occasionally null.

The question. *How close, in fact, is the read to the reference declared primary? And which other references are equally close, or closer than the reported outcome would suggest?*

Answering this question entails moving from a categorical attribution — this read belongs to this species — to a probabilistic description of the available evidence. This is the substance of the development phase documented here.

The present report sets out the rationale adopted, the results obtained on mock communities of known composition and, in closing, the fundamental limitation that the investigation brought to light, together with the directions available for overcoming it. A descriptive register has been adopted throughout, free of implementation detail, giving precedence to the statistical reasoning and its biological interpretation.

2 The rationale, in four steps

2.1 Step 1 — Making the alternatives visible

The first decision was epistemic before it was methodological: **to retain the alternatives rather than discard them**. Under its ordinary configuration the aligner reports only the best correspondence for each read; secondary correspondences are suppressed for economy of computation and storage.

Recovering them changes the nature of the elementary datum: a single attribution gives way to a **set of competing hypotheses**, each accompanied by its own alignment score. In the data examined here this set comprises, on average, ten candidate sequences per read (Figure 1a): ten references that the algorithm judges compatible with the observed read, of which only one survived into the final result.

This choice carries a cost — the alignment archive grows by approximately one fifth — and is therefore enabled at the operator's discretion, the ordinary procedure remaining unchanged when it is not requested.

2.2 Step 2 — Weighting the alternatives

Possessing ten candidates resolves nothing unless one establishes **how far** each is plausible. The alignment score supplies the necessary information: it increases with the number of matching bases and decreases with observed differences, so that the score difference between two candidates is proportional to the number of positions at which they diverge with respect to the read.

A transformation was adopted that converts the scores into **relative likelihood weights**, normalised to unit sum within each read. The constant governing the severity of this transformation was fixed at the

cost of a single nucleotide difference: a candidate diverging by one additional position relative to the best receives a weight of approximately one third. This calibration is deliberately **conservative** — a formal analysis of the scoring system would suggest a severity approximately threefold greater — in the sense that the method reports more residual ambiguity than the scores strictly warrant.

The adequacy of this choice was verified *a posteriori* by cross-validation, holding out each community in turn and optimising the constant on the remainder (§4.1): the optimised calibration fails to generalise, and the conservative value adopted proves preferable.

The score margin between the first and second choice is itself an informative quantity: its distribution (Figure 1b) clearly separates the taxonomic units genuinely present in the mock communities from spurious ones (median 60 versus 30). This observation becomes central in §5.

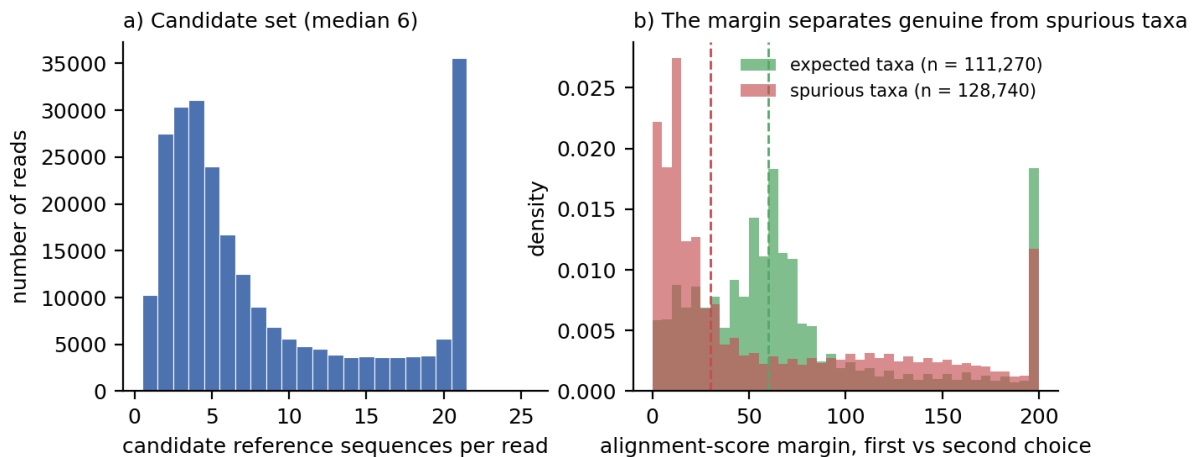


Figure 1: **Figure 1.** (a) Distribution of the number of candidate sequences per read after recovery of secondary correspondences. (b) Distribution of the alignment-score margin between first and second choice, distinguished between taxonomic units expected in the mock communities and spurious units; dashed lines denote medians.

2.3 Step 3 — Establishing the taxonomic depth the evidence supports

Weighting the candidates permits a finer question than “which species?”: **at which taxonomic rank is the evidence sufficient?**

For each read the taxonomic hierarchy is ascended — from species towards genus, family and order — and the deepest rank is identified at which a single taxon accumulates at least 90% of the total candidate weight. A read whose candidates are two species of the same genus is not declared ambiguous: it is declared **certain at genus level and indeterminate at species level**, which is the correct information.

Two methodological refinements carry direct biological consequence:

- The consensus threshold is referred solely to the mass **actually annotated** at that rank. A reference sequence lacking a specific epithet constitutes missing information, not disagreement; charging it as the latter would transfer the incompleteness of the archive onto the sample under analysis, attributing to the biological material an ambiguity that in fact belongs to the knowledge base.
- Taxonomic ranks are **ordinal** quantities and do not admit averaging. The typical resolution of a unit is therefore reported as the modal rank, accompanied by the full distribution of depths; an average would yield ranks that no read supports.

The consensus criterion adopted is conceptually cognate with the bootstrap threshold employed in Bayesian taxonomic classifiers (Wang et al. 2007), and its severity was likewise subjected to verification (§4.1).

2.4 Step 4 — Reapportioning ambiguous reads by likelihood

The final step addresses the quantitative consequence of the problem. If ten reads are compatible with two indistinguishable species, attributing all of them to whichever won a marginal contest introduces a systematic bias, the more severe the more the archive contains redundant sequences.

Reapportionment by **maximum likelihood** was adopted, using the expectation–maximisation algorithm (Dempster et al. 1977), an approach well established in transcript quantification (Li and Dewey 2011; Bray et al. 2016) and transferred to metagenomics (Lu et al. 2017) and, more recently, to long-read amplicons (Curry et al. 2022). The principle is that reads attributable unambiguously to a single reference — the only genuinely informative ones — determine the relative abundances, which in turn govern the apportionment of the ambiguous reads.

One aspect required particular care. If the procedure is initialised from the primary counts, a reference that never wins a contest begins at zero abundance and **can never subsequently acquire any**: its exclusion is self-confirming at every iteration. Since this is precisely the condition of indistinguishable sequences — those for which reapportionment exists in the first place — a uniform initialisation was adopted, in conformity with established practice in the field (Li and Dewey 2011; Bray et al. 2016). For two UNITE accessions identical over the amplified region, primary counts of 33 and 27 become 30.0 and 30.0: the arbitrary apportionment performed by the aligner is removed rather than inherited.

3 Results on mock communities

3.1 Validation design

Six mock yeast communities of known composition were employed (four to six species each, at defined proportions), amplified for the ITS region and sequenced on the Oxford Nanopore platform (Conti et al. 2023); the set comprises 18 samples and approximately 250,000 reads. Each estimate was compared against the expected composition, computing precision (the proportion of recovered taxa genuinely present), recall (the proportion of expected taxa genuinely recovered) and their harmonic mean F1, at a relative-abundance threshold of 0.05%.

3.2 Outcome

Estimator	Precision	Recall	F1	False positives per community
Primary counts	0.179	0.539	0.262	12.8
Likelihood-based reapportionment	0.227	0.539	0.316	9.0

The improvement in F1 amounts to **21%** and is attributable entirely to precision: recall is unchanged, whereas spurious taxa decline by almost one third (Figure 2). The gain is consistent across all six communities and derives from no parameter optimisation whatsoever (§4.1).

What does **not** improve warrants equal emphasis. Measures of agreement between observed and expected abundances — Pearson correlation and Bray–Curtis dissimilarity — remain unchanged (0.396 and 0.686

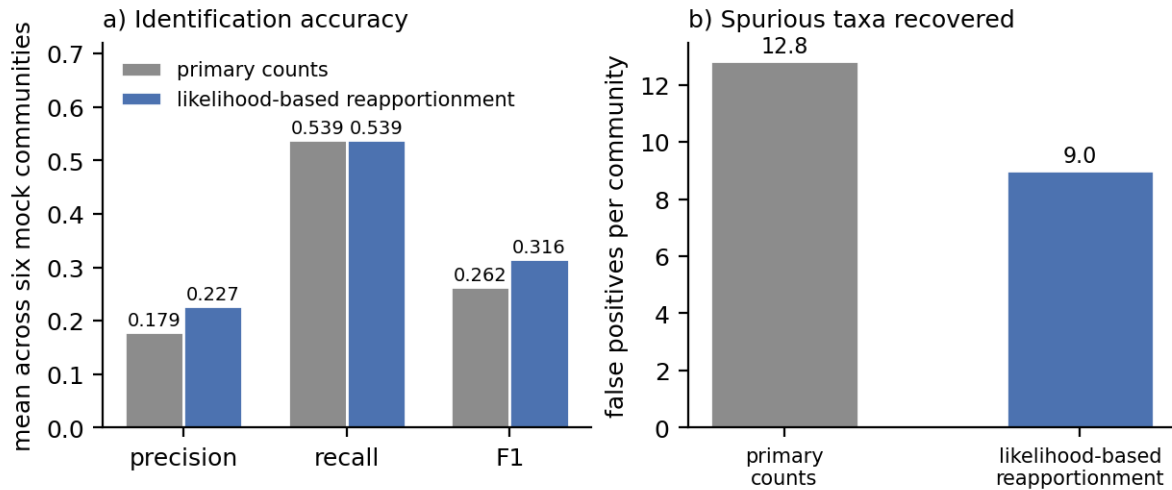


Figure 2: **Figure 2.** (a) Mean precision, recall and F1 across the six mock communities at a relative-abundance threshold of 0.05%. (b) Mean number of spurious taxa recovered per community.

respectively). The procedure therefore corrects *which* taxa are declared present, not *in what proportion*. This qualification is necessary: the benefit concerns the qualitative composition of the community, not its quantitative structure.

3.3 Parameter calibration and its verification

Three quantities govern the procedure: the severity of candidate weighting, the taxonomic consensus threshold and an optional regularisation of rare abundances. All three were subjected to systematic exploration (28 combinations) under leave-one-community-out cross-validation.

The outcome was conservative and, in its way, reassuring:

- optimised calibration **fails to generalise**: selecting parameters on five communities and applying them to the sixth yields results inferior to the conservative values adopted (F1 0.312 versus 0.316), and the winning combination varies from one hold-out to the next;
- regularisation of rare abundances degrades accuracy at every level of severity;
- the consensus threshold exerts no discriminant effect across the interval explored;
- the information contained in the scores is collectively worth a substantial increment: nullifying it — that is, treating all candidates as equivalent — reduces performance below that of simple primary counting.

The original values were therefore retained, but their adoption now rests upon statistical verification rather than upon convenience.

4 The fundamental limitation: when the closest sequence is not the correct one

4.1 The observation

One of the six communities — composed of four congeneric species of the genera *Debaryomyces* and *Dipodascus* — yielded a qualitatively inverted outcome: 87% of reads attributed to two species **absent** from the community, with the four expected species accounting collectively for 3%.

The instructive aspect is not the error but the array of assurances accompanying it. The two erroneous assignments are resolved at species level for 93% and 99.7% of reads respectively, with consensus support of 1.00 — the maximum attainable. Applying likelihood-based reappportionment does not correct the error: it accentuates it, from 87% to 93%. Credibility intervals obtained by resampling the reads declare with 95% confidence that the dominant expected species — 40% of the community in reality — is present at 0.00%, with bounds [0.00%; 0.00%]. The empirical coverage of such intervals, measured across all communities, is 4% against a nominal value of 95%.

4.2 The explanation

All the instruments of control described in Steps 2–4 share one property: they are **relative** measures, computed within the candidate set.

Instrument	Quantity measured	Reason for failure
Taxonomic consensus	disagreement <i>among</i> candidates	if all candidates are erroneous but one prevails clearly, there is no disagreement to detect
Likelihood-based reappportionment	<i>relative</i> apportionment of mass	it normalises over a set that does not contain the correct answer: the probabilities still sum to unity
Read resampling	sampling variability	with hundreds of thousands of reads this is negligible; the error here is epistemic in nature

What is absent is an **absolute** measure: how well the best available hypothesis in fact explains the data. The question “and what if none of these references were the correct one?” is posed by none of the three.

The phenomenon has recognised biological roots. The ITS marker exhibits non-negligible intraspecific variability and, in certain clades, insufficient interspecific separation (Nilsson et al. 2008; Estensmo et al. 2021); coverage of reference archives remains partial, to the point that the UNITE community itself has coined the notion of *dark taxa* for entities not referable to a described species (Nilsson et al. 2019; Abarenkov et al. 2023). The accuracy of molecular fungal identification is, for these reasons, structurally bounded (Lücking et al. 2020).

5 The controlled-exclusion experiment

5.1 Design

Since the critical condition — absence of the correct reference — can be **produced deliberately**, an experiment was designed that realises it against known truth. For each community and each expected species, all sequences of that species are removed from the archive and the analysis repeated, observing where the orphaned reads migrate and whether any indicator registers their departure. Thirty configurations were examined; one case was repeated with full realignment against the reduced archive, as a validity control on the rapid procedure.

5.2 First outcome: half the cases are not interrogable

In 15 configurations out of 30 the species to be removed **already receives no reads** in the ordinary analysis: its reads lie upon indistinguishable relatives. The phenomenon encompasses the entire congeneric community described above and several species of the genus *Saccharomyces*. The experiment can measure the loss of what one possesses, not of what one never possessed — and this asymmetry is itself a result: **half of the expected species already reside in a condition of silent irresolvability**.

5.3 Second outcome: the predictor of detectability is not genetic distance

For each of the 15 interrogable cases, two quantities were measured: the additional divergence induced by removal (the increase in distance between read and best residual reference) and the change in score margin between first and second choice, the latter interpretable as an alarm signal.

Contrary to expectation, genetic divergence does **not** predict detectability (Spearman's $\rho = +0.34$; $P = 0.22$). The significant predictor is the **concentration of the destination**, that is, what proportion of the orphaned reads converges upon a single receiving taxon ($\rho = -0.60$; $P = 0.019$). The separation is unequivocal:

Structure of the receiving neighbourhood	Cases	Detected	Median change in margin
Dispersed destination (< 70% on a single taxon)	7	6	+47% (the margin contracts: alarm)
Single destination ($\geq 70\%$ on a single taxon)	8	0	−185% (the margin expands)

The second finding merits explicit statement: where the missing species possesses a single well-represented congener, the reads converge upon it *en bloc* and the erroneous assignment appears **more secure than before**. The error not merely escapes detection: it actively conceals itself.

In all fifteen cases, moreover, the taxonomic consensus registers no change whatever (support between 0.98 and 1.00 following removal of the correct reference).

5.4 Biological interpretation

The picture that emerges is coherent and admits a natural reading in terms of the phylogenetic structure of the reference archive. What renders an absence perceptible is not the evolutionary remoteness of the missing species but the **configuration of its neighbourhood**:

- where the neighbourhood is **degenerate** — a cloud of approximately equidistant relatives — the orphaned reads disperse, no reference claims them with authority, and the score margin collapses: the absence leaves a measurable trace;
- where the neighbourhood contains a **single clean substitute**, the reads migrate to it *en bloc* and every statistical indicator improves: the absence leaves no trace at all.

A corollary of practical significance follows. In the congeneric cases 85–99% of the mass migrates nonetheless **within the correct genus**: the reconstruction remains faithful at genus level and unreliable at species level. Reporting the outcome at the rank actually supported by the data would remedy the majority of these cases without requiring any additional detection capability.

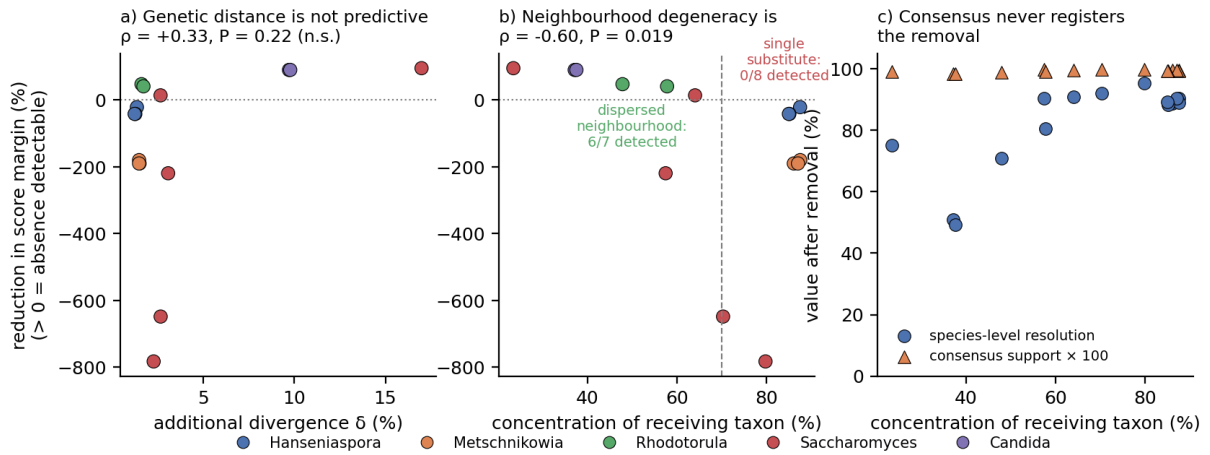


Figure 3: **Figure 3.** Outcome of the controlled-exclusion experiment (15 interrogable configurations). (a) Additional genetic divergence does not predict detectability. (b) Concentration of the destination does: with a degenerate neighbourhood the absence is perceptible, with a single congener it never is. (c) Taxonomic consensus remains unchanged in every configuration. Colours denote genera.

5.5 Third outcome: a useful indicator, within its own domain

The score margin between first and second choice is reflected in a quantity the aligner already supplies, the mapping quality (Li 2018) — the two are strongly associated ($\rho = 0.77$; $P < 0.001$). Employed as a discriminant between authentic and spurious taxonomic units and weighted by abundance, this quantity attains an area under the ROC curve of 0.97, against 0.62–0.78 for the consensus measures.

In the light of the exclusion experiment, however, this result must be **circumscribed**: the indicator intercepts cases with a degenerate neighbourhood — which include the two dominant false positives of the congeneric community — but not substitution by a single congener. It has accordingly been incorporated into the analytical report as a **screening instrument**, accompanied by explicit statement of its domain of validity, and not as an automatic filter.

6 Limitations

Primary limitation — congeneric substitution is not detectable. Where the archive contains a single well-represented close relative, none of the quantities presently computed signals the absence of the correct species; on the contrary, the indicators improve. Within the material examined, the domain of blindness comprises all cases with a concentrated destination, irrespective of genetic distance.

Confidence and correctness are distinct quantities. Measures of resolution and support quantify *concordance among the available references*, not their correctness. A maximal support value is compatible with a wholly erroneous attribution; this property is structural and must be declared in any use made of the results.

Current uncertainty estimates are not calibrated. Intervals obtained by resampling reads exhibit an empirical coverage of 4% against a nominal 95%, since they measure sampling variability alone and not the uncertainty arising from archive incompleteness. Their use in this form is not recommended.

The correction concerns presence, not proportion. The gain documented in §3 pertains to precision and richness; agreement between observed and expected abundances is unchanged.

Archive redundancy. Reference sequences are clustered into species hypotheses (Köljalg et al. 2013), yet a single species may appear under many distinct hypotheses: in the material examined *Saccharomyces cerevisiae* is distributed across 31 units. Reapportionment operates upon the individual reference and does not merge them; aggregation by taxonomic label is required for richness analysis.

Extent of validation. The results derive from six mock communities, a single marker, a single reference archive and a single sequencing chemistry. The quantities reported — in particular the area under the ROC curve of the screening indicator — are measured upon the same material from which the indicator was identified, and await independent confirmation.

A richer statistical model did not improve the outcome. A complete probabilistic model was implemented and evaluated, incorporating an error model estimated from the data and explicit treatment of chimeric reads. Upon this material it proves **less accurate** than the simpler procedure (F1 0.134 versus 0.316), because a likelihood computed position by position discriminates less effectively than the overall alignment score, which embodies the aligner’s complete optimisation. The finding was verified by excluding three alternative explanations.

7 Prospective improvements

The directions below are ordered by the ratio of expected value to implementation burden. The first two carry no methodological risk; the third constitutes the innovative core.

(i) Absolute plausibility assessment for each read. Rather than comparing candidates against one another, to ask whether the alignment observed with the best of them is compatible with the sequencer’s error rate, estimated from the data themselves. Divergence exceeding that explicable by instrumental error would signal that the correct reference may be absent. The quantity aggregated by taxonomic unit — the *proportion of poorly explained reads* — would be the system’s first absolute measure. The exclusion experiment already delineates its expected domain: power within degenerate neighbourhoods, reduced power under congeneric substitution.

(ii) Reporting the outcome at the rank actually supported. Where two references are indistinguishable, the individual abundances are not identifiable but their sum is. Reporting the taxon at the finest taxonomic

rank supported by the data would convert a silent error into a declared property and would remedy, as shown in §5.3, the majority of congeneric cases.

(iii) Estimating the unrepresented community fraction. To extend the model with an explicit *unknown source* component, whose likelihood is constructed by marginalising over the distribution of congeneric divergences — a distribution estimable from the archive itself, and already the object of the companion reference-database analysis tool developed within the same doctoral project. The resulting parameter would furnish, for each sample, an estimate of the fraction of the community that the archive is unable to explain: a quantity no instrument presently reports and which ecological analysis requires. The principal risk is statistical in nature — the possible non-identifiability between this parameter and the error rate, both compatible with elevated divergences — and requires formal verification before proceeding.

(iv) Uncertainty referred to the archive rather than to the reads. To replace read resampling with repeated perturbations of the reference archive: the stability of abundances under variation of the reference set is the form of uncertainty pertinent to this problem, and would yield intervals that widen where the attribution is fragile.

(v) Extension of the validation. To complete the exclusion experiment with full realignment across a grid of distances, and to replicate the entire design upon a second marker, a second kingdom and a second sequencing chemistry.

A final consideration, which we regard as the most consequential interpretatively. The exclusion experiment confirms by independent means that **no statistical correction applied downstream recovers information the marker does not contain**: in the congeneric community, reapportionment degrades the outcome, whereas use of a more extended marker resolves it. The improvement documented in this phase acts upon the inference; the residual limitation belongs to the molecular design and to the completeness of the archive, and must be addressed there.

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